Nonexistence of Strongly Regular Graphs via Gegenbauer Positivity and SRG Counting

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Abstract

This paper develops a natural extension of the multipoint spherical semidefinite method of Liao and Yu [LY26], studying the interaction between semidefinite programming and strongly regular graph counting. Starting from the positive semidefinite kernels furnished by Gegenbauer polynomials, we express weighted kernel sums and counting identities in the same labelled induced-subgraph counts. Eliminating the counting equations gives a semidefinite feasibility problem. Numerical dual solutions guide the construction of rational certificates, whose contradiction identities are verified exactly. Our aim is a systematic method that treats many parameter sets through the same construction. For least eigenvalue -3, it gives new counting certificates for $(351, 140, 73, 44)$, $(550, 162, 75, 36)$, $(703, 182, 81, 35)$ and $(1344, 221, 88, 26)$, previously excluded by Liao and Yu [LY26, Theorem 1], and for six additional parameter sets:

$$(375, 102, 45, 21), \quad (405, 132, 63, 33), \quad (441, 88, 35, 13),$$
$$(476, 133, 60, 28), \quad (540, 147, 66, 30), \quad (575, 112, 45, 16).$$

Seven certificates use Gegenbauer positivity and counting information alone at the displayed orders. The cases $v = 375, 441, 575$ additionally use proved short-vector restrictions, unique vertex extensions, and integer common-neighbour inequalities. The proofs end in displayed inequalities with incompatible signs. Their justification is given by the dual identities and the proved positivity and counting relations; accompanying code automates the longer rational arithmetic.

1 Introduction

A strongly regular graph with parameters (v, k, λ, μ) is a k-regular graph on v vertices in which adjacent vertices have λ common neighbours and distinct nonadjacent vertices have μ common neighbours. Its eigenspaces provide spherical two-distance sets, whose sizes can be bounded by semidefinite programming. The multipoint spherical approach of Liao and Yu [LY26] obtains nonexistence results by bounding the size of such sets. Here our goal also includes understanding how semidefinite programming interacts with the counting structure of strongly regular graphs. We start directly from Gegenbauer positivity and formulate feasibility in induced-subgraph counts. The eigenspace embedding supplies the inner products at which the kernels are evaluated; a cardinality bound for general spherical two-distance sets is no longer the optimization problem. This formulation retains the degree and common-neighbour information of the graph.

Kováčiková already combined Gegenbauer inequalities with induced-subgraph counts [K15, Proposition 4.2, Lemma 4.2, and Sections 4.2–4.3]. Her construction represents induced subgraphs by normalized nonzero sums of their spherical vertex vectors and groups the resulting sums by induced union types. Here we use Musin's projected kernels at fixed reference configurations, with arbitrary weights indexed by external adjacency profiles. The resulting positive semidefinite matrices are linear in labelled induced-subgraph counts. We combine these matrices with the SRG extension equations and, where needed, further proved structural relations to construct explicit rational nonexistence certificates.

Our four-point experiments in late 2025 combined the counting relations of [K15, Proposition 3.2 and Section 3.3.3] with semidefinite constraints but excluded none of the tested parameter sets. The certificates below use counts through five, six, or seven vertices.

Musin's positivity theorem [M08] supplies inequalities in the distribution of small induced subgraphs, while double counting supplies linear equations in the same variables. Eliminating the latter gives a semidefinite problem whose dual guides the search for rational contradiction identities. Our aim is a reusable method that treats many parameter sets through one construction. The counting identities, kernel inequalities, and dual certificate mechanism are shared across the applications; further structural information enters through the same count variables. The certificate takes the form of a rational identity in graph counts, and each case ends in an explicit contradictory inequality. The displayed formulas and the proved relations establish the argument; the verification programs check its finite coefficient calculations. For example, the reported six-point spherical upper bound for $(540, 147, 66, 30)$ is 548 [LY26, Table 3]; here five-point counts, three reference vertices, and degrees $1, 2$ give a contradiction.

The focus on least eigenvalue -3 is guided both by the candidate list of Koolen and Gebremichel [KG22, Table 1] and by our repeated computational experiments with spherical bounds and SRG counting. Their 2022 Table 1 lists twelve unresolved candidates among primitive strongly regular graphs with least eigenvalue -3, $v > 276$, and $\mu \notin \{6, 9\}$. Liao and Yu [LY26, Theorem 1] subsequently excluded the four rows marked † in Table 1. The ten parameter sets proved here are listed in that table; our aim is to exhibit their nonexistence within a common semidefinite framework.

Table 1: The ten excluded parameter sets. Here r has multiplicity d, and p is the largest graph order in the certificate. † marks the four cases previously treated by Liao–Yu [LY26]. Every certificate combines Gegenbauer positivity with strongly regular graph counting information. The final column records whether the additional structural information proved in Section 4 is used.

v	k	λ	μ	r	d	p	Additional structural information
351†	140	73	44	32	26	6	Not needed
375	102	45	21	27	34	7	Used (Section 4)
405	132	63	33	33	30	7	Not needed
441	88	35	13	25	44	7	Used (Section 4)
476	133	60	28	35	34	6	Not needed
540	147	66	30	39	35	5	Not needed
550†	162	75	36	42	33	5	Not needed
575	112	45	16	32	46	7	Used (Section 4)
703†	182	81	35	49	37	5	Not needed
1344†	221	88	26	65	56	5	Not needed

Theorem 1.1. *No strongly regular graph exists for any parameter set in Table 1.*

Seven cases are proved in Section 3 from Gegenbauer positivity and counting information. For 375, 441, and 575, Section 4 adds structural information to the same method. The displayed orders record the available certificates; minimal orders are not asserted.

Data and code availability. The article, supplementary material, exact certificates, and verification programs are available from the public companion repository. The certificate data used in the proofs are identified by the fixed repository snapshot `85cd39b` [Data]. Appendix A.2 gives the certificate locations and the command for reproducing their exact arithmetic checks.

Section 2 develops the construction and its Lagrange dual. Sections 3 and 4 give the contradictions. Detailed rational coefficients and verification conventions are collected in the appendices and the separate 405 certificate supplement [S405]. Appendix A.2 identifies the accompanying certificate snapshot, its exact inputs, and the common verification command.

2 Gegenbauer positivity and counting information

We express weighted Gegenbauer kernel sums in labelled induced-subgraph counts and derive exact rational nonexistence certificates.

2.1 Vertices, spherical vectors, and fixed reference vertices

We use the normalized eigenspace embedding in [LY26, Section 4.1], specialized to least eigenvalue -3. Let $\Gamma = (V, E)$ be a primitive strongly regular graph with parameters (v, k, λ, μ), adjacency matrix $\mathrm{Adj}(\Gamma)$, and nontrivial eigenvalues $r, -3$, with r of multiplicity d. All parameter sets considered satisfy $\mu > 0$ and $v - 2k + \lambda > 0$, which force the graph and its complement to be connected. Then

$$r = \lambda - \mu + 3, \qquad k = 3r + \mu, \qquad d = \frac{3(v-1) - k}{r+3}. \tag{2.1}$$

These formulas follow from the quadratic equation for the nontrivial eigenvalues and $\operatorname{tr}\mathrm{Adj}(\Gamma) = 0$. The orthogonal projector onto the r-eigenspace is

$$E_r = \frac{1}{r+3}\left(\mathrm{Adj}(\Gamma) + 3I - \frac{k+3}{v}J\right).$$

Here I and J are the $v \times v$ identity and all-one matrices. The projector has diagonal d/v. Normalizing its columns gives unit vectors $p_x \in S^{d-1} \subset \mathbb{R}^d$, one for each $x \in V$, with

$$\langle p_x, p_y\rangle = \begin{cases} 1, & x = y, \\ \beta, & x \sim_\Gamma y, \\ \alpha, & x \neq y,\ x \not\sim_\Gamma y, \end{cases} \qquad \alpha = -\frac{k+3}{3v-k-3}, \quad \beta = \frac{v-k-3}{3v-k-3}. \tag{2.2}$$

Equivalently, $\beta = r/k$ and $\alpha = -(r+1)/(v-k-1)$.

Fix distinct vertices $r_0, \dots, r_{m-1}$ whose vectors are linearly independent, with $1 \le m \le d-2$, and put

$$W = \operatorname{span}\{p_{r_0}, \dots, p_{r_{m-1}}\}, \qquad D = d - m, \qquad X = V \setminus \{r_0, \dots, r_{m-1}\}.$$

For $x \in X$, write $p_x = p_x^{\parallel} + p_x^{\perp}$ for its orthogonal decomposition in $W \oplus W^{\perp}$.

2.2 The scalar kernel and its positivity

For $\ell \in \mathbb{Z}_{\ge 0}$, let $P_\ell^{(D)}$ be the Gegenbauer polynomial for S^{D-1}, normalized by $P_\ell^{(D)}(1) = 1$. For $D = 2$ use $P_\ell^{(2)}(\cos\theta) = \cos(\ell\theta)$. Define

$$K_{\ell,W}(x,y) = (\|p_x^{\perp}\|\|p_y^{\perp}\|)^{\ell} P_\ell^{(D)}\left(\frac{\langle p_x^{\perp}, p_y^{\perp}\rangle}{\|p_x^{\perp}\|\|p_y^{\perp}\|}\right). \tag{2.3}$$

At a zero orthogonal component use its polynomial extension: $K_{0,W} = 1$, and $K_{\ell,W}(x,y) = 0$ for $\ell \geq 1$ if either component is zero. Musin's construction and positivity theorem [M08, Definition 2.1 and Theorem 3.1] give

$$\sum_{x,y\in X} a_x a_y K_{\ell,W}(x,y) \geq 0 \qquad ((a_x)_{x\in X} \in \mathbb{R}^X). \tag{2.4}$$

Thus Gegenbauer positivity means that the kernel matrix $(K_{\ell,W}(x,y))_{x,y\in X}$ is positive semidefinite. Every double sum uses ordered pairs and includes $x = y$. Let $u_x \in \mathbb{R}^m$ be the coordinates of $p_x^{\parallel}$ in an orthonormal basis of W, and put $t_{xy} = \langle p_x, p_y\rangle$. Musin's variables then satisfy $1 - \|u_x\|^2 = \|p_x^{\perp}\|^2$ and $t_{xy} - u_x^T u_y = \langle p_x^{\perp}, p_y^{\perp}\rangle$; this gives exactly (2.3).

In degrees zero, one, and two the kernels are

$$K_{0,W} = 1, \quad K_{1,W}(x,y) = \langle p_x^{\perp}, p_y^{\perp}\rangle, \quad K_{2,W}(x,y) = \frac{D\langle p_x^{\perp}, p_y^{\perp}\rangle^2 - \|p_x^{\perp}\|^2\|p_y^{\perp}\|^2}{D-1}. \tag{2.5}$$

Writing $q_x = p_x^{\perp}$ only in the next display, their sums are

$$\begin{aligned}
\sum_{x,y} a_x a_y K_{0,W}(x,y) &= \left(\sum_x a_x\right)^2, \\
\sum_{x,y} a_x a_y K_{1,W}(x,y) &= \left\|\sum_x a_x q_x\right\|^2, \\
\sum_{x,y} a_x a_y K_{2,W}(x,y) &= \frac{D}{D-1}\left\|\sum_x a_x \left(q_x q_x^T - \frac{\|q_x\|^2}{D} I_D\right)\right\|_F^2.
\end{aligned} \tag{2.6}$$

The last equality follows from $\langle uu^T, vv^T\rangle_F = \langle u, v\rangle^2$ by taking trace-free parts. Thus these three degrees also have a direct proof of positivity.

2.3 Labelled counts and one-vertex extensions

All graphs are finite and simple. A labelled t-vertex graph F has vertex set $[t] = \{0, \dots, t-1\}$. Define

$$N_F = \#\{\theta : [t] \hookrightarrow V : i \sim_F j \iff \theta(i) \sim_\Gamma \theta(j) \text{ for } i \neq j\}. \tag{2.7}$$

A hooked arrow between graphs denotes an induced embedding. In particular, $N_{K_1} = v$, $N_{K_2} = vk$, and $N_{K_3} = vk\lambda$. Let $\mathcal{G}_t$ contain one labelled representative of each isomorphism class on $[t]$. Isomorphic graphs have equal counts, and each vertex subset inducing F contributes $|\operatorname{Aut} F|$ embeddings. Hence

$$\sum_{F\in\mathcal{G}_t} \frac{N_F}{|\operatorname{Aut} F|} = \binom{v}{t}. \tag{2.8}$$

For H on $[h]$ and $B \subseteq [h]$, let H_B be the graph on $[h+1]$ obtained by making the new vertex h adjacent exactly to B. For fixed $B_0 \subseteq [h]$ with $|B_0| \leq 2$, put

$$e(H, B_0) = \begin{cases} v - h, & B_0 = \varnothing, \\ k - \deg_H(i), & B_0 = \{i\}, \\ \lambda - |\mathcal{N}_H(i) \cap \mathcal{N}_H(j)|, & B_0 = \{i,j\},\ i \sim_H j, \\ \mu - |\mathcal{N}_H(i) \cap \mathcal{N}_H(j)|, & B_0 = \{i,j\},\ i \not\sim_H j. \end{cases} \tag{2.9}$$

Here $\mathcal{N}_H(i)$ denotes the neighbourhood of i in H.

Proposition 2.1. *For every H, B_0 as above,*

$$e(H, B_0) N_H = \sum_{B_0 \subseteq B \subseteq [h]} N_{H_B}. \tag{2.10}$$

Proof. Count pairs (φ, x) where $\varphi : H \hookrightarrow \Gamma$, $x \notin \varphi([h])$, and x is adjacent to every $\varphi(i)$ with $i \in B_0$. For each φ there are $e(H, B_0)$ choices. Alternatively set $B = \{i : x \sim_\Gamma \varphi(i)\}$. Extending φ by $h \mapsto x$ gives an induced embedding of exactly one H_B, with $B \supseteq B_0$. Restriction reverses this construction. □

Isomorphic extensions occur with their full multiplicities; no automorphism factor appears. This is the labelled form of the extension counts in [K15, Proposition 3.2(i)–(iv)]. For example,

$$(\lambda - 1)N_{K_3} = N_{K_4 - e} + N_{K_4}.$$

2.4 Weights indexed by adjacency sets

Fix H on $[m]$, where $1 \le m \le d-2$, and form its Gram matrix

$$G_H = (1-\alpha)I_m + \alpha J_m + (\beta - \alpha)\operatorname{Adj}(H). \tag{2.11}$$

Assume $G_H \succ 0$, so every induced copy of H has independent spherical vectors. For $\varphi : H \hookrightarrow \Gamma$, set

$$W_\varphi = \operatorname{span}\{p_{\varphi(0)}, \dots, p_{\varphi(m-1)}\}, \qquad K_\ell^\varphi = K_{\ell, W_\varphi}.$$

For an induced embedding $\iota : H \hookrightarrow G$, define

$$\sigma_\iota(z) = \{i \in [m] : z \sim_G \iota(i)\}, \qquad z \in V(G) \setminus \iota([m]). \tag{2.12}$$

Let $\mathbf{1}_B$ be the characteristic column of $B \subseteq [m]$. Order $2^{[m]}$ lexicographically by these columns. For $m = 3$ the order is

$$\varnothing, \{2\}, \{1\}, \{1,2\}, \{0\}, \{0,2\}, \{0,1\}, \{0,1,2\}. \tag{2.13}$$

Choose any $f : 2^{[m]} \to \mathbb{R}$, independently of φ, and identify it with its column of 2^m values. Apply (2.4) with $a_x = f(\sigma_\varphi(x))$ and then sum over φ:

$$S(H, \ell, f) := \sum_{\varphi : H \hookrightarrow \Gamma} \sum_{x,y \in V \setminus \varphi([m])} f(\sigma_\varphi(x)) f(\sigma_\varphi(y)) K_\ell^\varphi(x,y) \ge 0. \tag{2.14}$$

There is no division by $|\operatorname{Aut} H|$; an empty sum is zero. Each summand has support $\varphi([m]) \cup \{x, y\}$, of size $m+1$ if $x = y$ and $m+2$ otherwise.

2.5 Exact coefficients on abstract graphs

Put

$$b(B) = \alpha \mathbf{1} + (\beta - \alpha)\mathbf{1}_B \in \mathbb{R}^m,$$

where $\mathbf{1}$ is the all-one column. For $\psi : H \hookrightarrow F$ and $a, b \in V(F) \setminus \psi([m])$, write $b_a^\psi = b(\sigma_\psi(a))$ and let t_{ab} be $1, \beta, \alpha$ according as $a = b$, $a \sim_F b$, or $a \ne b$ and $a \not\sim_F b$. If $P_\ell^{(D)}(t) = \sum_{j=0}^{\lfloor \ell/2 \rfloor} \gamma_{\ell,j}^{(D)} t^{\ell - 2j}$, define

$$K_\ell^\psi(a,b) = \sum_{j=0}^{\lfloor \ell/2 \rfloor} \gamma_{\ell,j}^{(D)} \left(t_{ab} - (b_a^\psi)^T G_H^{-1} b_b^\psi\right)^{\ell - 2j} \left[(1 - (b_a^\psi)^T G_H^{-1} b_a^\psi)(1 - (b_b^\psi)^T G_H^{-1} b_b^\psi)\right]^j. \tag{2.15}$$

This polynomial is defined even for unrealizable Gram data. For an actual copy, orthogonal projection gives

$$\|p_x^\perp\|^2 = 1 - b(\sigma_\varphi(x))^T G_H^{-1} b(\sigma_\varphi(x)), \quad \langle p_x^\perp, p_y^\perp \rangle = \langle p_x, p_y \rangle - b(\sigma_\varphi(x))^T G_H^{-1} b(\sigma_\varphi(y)),$$

so (2.15) agrees with (2.3); see [M08, Corollary 3.2] and [KY22, Definition 2.11]. Positivity applies to the sums on actual vectors, not to individual formal coefficients.

Define

$$c_{H,\ell,f}(F) = \frac{1}{|\operatorname{Aut} F|} \sum_{\psi: H \hookrightarrow F} \sum_{\substack{a,b \in V(F) \setminus \psi([m]) \\ \psi([m]) \cup \{a,b\} = V(F)}} f(\sigma_\psi(a)) f(\sigma_\psi(b)) K_\ell^\psi(a,b). \tag{2.16}$$

The inner sum uses the diagonal when $|F| = m+1$ and both ordered pairs of the two exterior vertices when $|F| = m+2$.

Proposition 2.2. *For the actual labelled counts,*

$$S(H,\ell,f) = \sum_{F \in \mathcal{G}_{m+1} \cup \mathcal{G}_{m+2}} c_{H,\ell,f}(F) N_F \geq 0. \tag{2.17}$$

Proof. Group the summands of (2.14) by their support U. If $\Gamma[U] \cong F$, choose one isomorphism $\theta : F \to \Gamma[U]$. Then $(\psi, a, b) \mapsto (\theta \circ \psi, \theta(a), \theta(b))$ is a bijection between the triples in (2.16) and those supported on U in (2.14). It preserves the profiles and kernels, without requiring invariance of f. Each U contributes $|\operatorname{Aut} F| c_{H,\ell,f}(F)$, and there are $N_F/|\operatorname{Aut} F|$ such subsets. □

2.6 The semidefinite model and counting elimination

Choose an integer counting order $1 \leq p \leq v$. Let $\mathcal{G}_{\leq p} = \bigcup_{t=1}^{p} \mathcal{G}_t$ and $N = (N_F)_{F \in \mathcal{G}_{\leq p}}$. Collect (2.10) for $1 \leq h < p$, $|B_0| \leq 2$ and (2.8) for $1 \leq t \leq p$ into

$$AN = b, \qquad \mathcal{P} = \{N \in \mathbb{R}^{\mathcal{G}_{\leq p}} : AN = b,\ N_F \geq 0\}. \tag{2.18}$$

Here b is the right-hand-side column of the counting equations. The matrix A has one column per graph type, with isomorphic extensions combined. Every actual SRG supplies a point of $\mathcal{P}$; integrality is relaxed. By (2.8),

$$0 \leq N_F \leq |\operatorname{Aut} F| \binom{v}{|F|}, \tag{2.19}$$

so $\mathcal{P}$ is compact, possibly empty.

Choose finitely many pairs (H,ℓ) with $H \in \mathcal{G}_m$, $1 \leq m \leq \min(p-2, d-2)$, $G_H \succ 0$, and $\ell \in \mathbb{Z}_{\geq 0}$. Since (2.16) is quadratic in f, there is a unique symmetric $2^m \times 2^m$ matrix $C_{H,\ell}(F)$ such that

$$f^T C_{H,\ell}(F) f = c_{H,\ell,f}(F).$$

Set $C_{H,\ell}(F) = 0$ at other graph orders, and put

$$M_{H,\ell}(N) = \sum_F N_F C_{H,\ell}(F), \qquad S(H,\ell,f;N) = f^T M_{H,\ell}(N) f. \tag{2.20}$$

For actual counts, $S(H,\ell,f;N) = S(H,\ell,f)$; the semicolon notation extends the same linear expression to formal count vectors. This adjacency-profile indexing is the two-distance, exterior-point form of the label-indexed multipoint constraints in [KY22, Theorem 2.13]. All weights at once give

$$\begin{aligned} &\text{minimize}_N && 0 \\ &\text{subject to} && AN = b, \quad N_F \geq 0 \quad (F \in \mathcal{G}_{\leq p}), \\ &&& M_{H,\ell}(N) \succeq 0 \quad \text{for each selected } (H,\ell). \end{aligned} \tag{2.21}$$

Every hypothetical graph gives a feasible point, so infeasibility excludes the parameters. A dual certificate supplies a fixed finite family of weights giving the contradiction.

Eliminating the counting equations. Row-reduce $[A \mid b]$, retaining any inconsistent row. If inconsistent, counting alone gives the contradiction. Otherwise let $q = |\mathcal{G}_{\le p}| - \operatorname{rank} A$ and choose the non-pivot counts $N_{F_1}, \dots, N_{F_q}$ as coordinates $\xi_1, \dots, \xi_q$. Then all solutions, and only those solutions, have the form

$$N_F(\xi) = a_{F0} + \sum_{j=1}^{q} a_{Fj}\xi_j, \qquad N_{F_j}(\xi) = \xi_j, \qquad \xi \in \mathbb{R}^q. \tag{2.22}$$

Indeed, after ordering pivot columns first, the reduced system is $(I\ R)(N_{\text{piv}}, N_{\text{free}})^T = c$; thus $N_{\text{piv}} = c - R\xi$ and $N_{\text{free}} = \xi$. Abbreviate $M_{H,\ell}(N(\xi))$ by $M_{H,\ell}(\xi)$, and, for $0 \le j \le q$, set

$$M_{H,\ell}^{(j)} = \sum_F a_{Fj} C_{H,\ell}(F), \qquad M_{H,\ell}(\xi) = M_{H,\ell}^{(0)} + \sum_{j=1}^{q} \xi_j M_{H,\ell}^{(j)}. \tag{2.23}$$

The reduced primal minimizes 0 over $\xi \in \mathbb{R}^q$ subject to $N_F(\xi) \ge 0$ for *every* F and $M_{H,\ell}(\xi) \succeq 0$ for every selected pair.

2.7 The Lagrange dual and exact certificates

Use the convention of Boyd and Vandenberghe [BV04, Sections 5.8.1 and 5.9.1]: write the inequalities as $-N_F(\xi) \le 0$ and $-M_{H,\ell}(\xi) \preceq 0$, with multipliers $z_F \ge 0$ and $Y_{H,\ell} \succeq 0$, where $Y_{H,\ell}$ has the same size as $M_{H,\ell}$. With the trace inner product $\langle Y, M\rangle = \operatorname{tr}(YM)$, the Lagrangian is

$$\begin{aligned} L(\xi; z, Y) &= -\sum_F z_F N_F(\xi) - \sum_{H,\ell} \langle Y_{H,\ell}, M_{H,\ell}(\xi)\rangle \\ &= -c_0(z, Y) - \sum_{j=1}^{q} \xi_j c_j(z, Y), \\ c_j(z, Y) &:= \sum_F z_F a_{Fj} + \sum_{H,\ell} \langle Y_{H,\ell}, M_{H,\ell}^{(j)}\rangle. \end{aligned} \tag{2.24}$$

The definition of c_j includes $j = 0$. The equalities are already imposed by (2.22); taking the infimum over $\mathbb{R}^q$ gives

$$g(z, Y) = \inf_{\xi} L(\xi; z, Y) = \begin{cases} -c_0(z, Y), & c_j(z, Y) = 0 \ (1 \le j \le q), \\ -\infty, & \text{otherwise.} \end{cases}$$

Consequently the dual is

$$\begin{array}{ll} \text{maximize}_{z,Y} & -c_0(z, Y) \\ \text{subject to} & c_j(z, Y) = 0 \quad (1 \le j \le q), \\ & z_F \ge 0, \quad Y_{H,\ell} \succeq 0. \end{array} \tag{2.25}$$

For a primal feasible ξ, positive semidefiniteness gives $g(z, Y) \le L(\xi; z, Y) \le 0$. A positive dual value therefore excludes a graph and, by scaling, can be normalized to 1. We seek

$$c_0(z, Y) = -1, \qquad c_j(z, Y) = 0 \ (1 \le j \le q), \qquad z_F \ge 0, \quad Y_{H,\ell} \succeq 0. \tag{2.26}$$

This sufficient certificate uses only weak duality; no strict feasibility or converse assertion about dual attainment is assumed.

From dual matrices to scalar identities. Index the rank-one terms of all positive semidefinite dual blocks by a single finite set of indices i, recording the reference graph H_i and degree ℓ_i of each term. Then

$$Y_{H,\ell} = \sum_{i:\,(H_i,\ell_i)=(H,\ell)} \eta_i f_i f_i^T, \qquad f_i \in \mathbb{R}^{2^{|H_i|}}, \quad \eta_i > 0.$$

Thus (2.26) becomes

$$\boxed{\sum_i \eta_i S(H_i,\ell_i,f_i;N(\xi)) + \sum_F z_F N_F(\xi) = -1 \qquad (\xi \in \mathbb{R}^q).} \tag{2.27}$$

For actual counts, every term on the left is nonnegative. The following sufficient criterion isolates the mathematical conclusion from the search.

Theorem 2.3 (Exact nonexistence certificate)**.** *Fix a parameter set and the counting and kernel constraints above. For finitely many admissible pairs (H_i,ℓ_i) and rational directions f_i, put $S_i(N) = S(H_i,\ell_i,f_i;N)$. Suppose there are rational multipliers $\eta_i, z_F \geq 0$ and a rational number $\varepsilon > 0$ such that*

$$\sum_i \eta_i S_i(N) + \sum_F z_F N_F = -\varepsilon \qquad \text{whenever } AN = b. \tag{2.28}$$

Then no strongly regular graph with the prescribed parameters exists.

The same conclusion holds if proved linear equalities are appended to $AN = b$ and the identity includes additional terms $\theta_a \mathcal{I}_a(N)$ on the left, where $\theta_a \in \mathbb{Q}_{\geq 0}$ and each rational affine expression $\mathcal{I}_a(N)$ is proved nonnegative for the counts of every hypothetical graph with these parameters. In that case the identity is required on the augmented affine solution space.

Proof. The counts of a hypothetical graph satisfy all imposed equalities. Each $S_i(N)$ is nonnegative by Proposition 2.2, each N_F is nonnegative, and each additional $\mathcal{I}_a(N)$ is nonnegative by hypothesis. The identity would give $0 \leq -\varepsilon < 0$. □

Count upper bounds, written as $U - x \geq 0$, are examples of the additional expressions. Section 4 also supplies linearized integer inequalities and additional counting equalities.

Recovering rational weights. Choose rational directions f_i and compute

$$\kappa_{ij} = f_i^T M^{(j)}_{H_i,\ell_i} f_i, \qquad S(H_i,\ell_i,f_i;N(\xi)) = \kappa_{i0} + \sum_{j=1}^{q} \kappa_{ij}\xi_j.$$

Then solve the rational linear feasibility problem

$$\eta_i, z_F \geq 0, \qquad \sum_i \eta_i \kappa_{ij} + \sum_F z_F a_{Fj} = \begin{cases} -1, & j = 0, \\ 0, & 1 \leq j \leq q. \end{cases} \tag{2.29}$$

A real solution of this rational linear system implies a rational one. Indeed, fix the coordinates that vanish at a real feasible point. The remaining rational equalities define a nonempty rational affine space by row reduction. Its rational points are dense, and sufficiently close rational points preserve positivity of the remaining coordinates. This implication assumes feasibility after the rational directions have been fixed: neither a numerical dual solution nor its rational approximation guarantees that feasibility. We retain only multipliers satisfying every equality and sign condition exactly; each block $\eta_i f_i f_i^T$ is then positive semidefinite by construction.

Verify the parametrization by

$$A(a_{F0})_F = b, \qquad A(a_{Fj})_F = 0 \ (1 \leq j \leq q), \qquad \operatorname{rank}(a_{Fj})_{F,1\leq j\leq q} = q = |\mathcal{G}_{\leq p}| - \operatorname{rank} A. \tag{2.30}$$

Together with (2.16) and (2.29), these checks verify the hypotheses of Theorem 2.3, independently of the numerical search.

3 Exact contradictions from the counting and kernel constraints

The spherical data are determined by (2.1)–(2.2). We give seven exact contradictions. The displayed functions and the coefficients in the appendices and Supplement [S405] specify each certificate completely.

3.1 Four five-point certificates

Use $p = 5$, $m = 3$, and degrees $\ell \in \{1, 2\}$. The 52 graph types on at most five vertices give a counting system of rank 48 for each of $v = 540, 550, 703, 1344$. Its four non-pivot columns, in lexicographic edge-string order, are

$$F_1 = K_5 \setminus P_3, \quad F_2 = K_5 \setminus 2K_2, \quad F_3 = K_5 \setminus K_2, \quad F_4 = K_5, \qquad \xi_j = N_{F_j}. \tag{3.1}$$

Here $K_5 \setminus P_3$ means deletion of the two edges of a three-vertex path. These are the original labelled counts, so $\xi_j \geq 0$. Their automorphism orders are $4, 8, 12, 120$, respectively. The corresponding ten-bit edge strings are 0011111111, 0111111011, 0111111111, and 1111111111. They use the edge order specified in Appendix A. The total-count equations in Section 2 may be retained: they already follow from the empty-set extension equations and $N_{K_1} = v$.

For three reference vertices, an edge string lists $01, 02, 12$. Thus 001 is one edge and an isolated vertex, 011 is a path, and 111 is a triangle; their automorphism orders are $2, 2, 6$. The functions in Table 2 are given by all eight values in exactly the order of Section 2:

$$\varnothing, \{2\}, \{1\}, \{1, 2\}, \{0\}, \{0, 2\}, \{0, 1\}, \{0, 1, 2\}.$$

Index the rows within each v-block of Table 2 starting at 1. A row displaying qf_i defines f_i by dividing every entry by the row's denominator q. All reference Gram matrices are positive definite. Indeed, in the four cases $2\max(|\alpha|, |\beta|) < 1$, so strict diagonal dominance proves this for every three-vertex graph.

Table 2: Explicit rational functions for the four five-point proofs. The last column supplies $t_i > 0$ in (3.2).

v	H_i	ℓ_i	q	qf_i	t_i
540	001	1	10	$(10, 6, 6, 0, 7, 0, 0, 0)$	102
	111	2	10	$(3, -4, -4, 10, -4, 10, 10, 0)$	80
	011	1	10	$(10, 4, 9, 0, 9, 0, 0, 0)$	61
	011	1	10	$(10, 5, 8, 0, 8, 0, 0, 0)$	37
	011	1	10	$(10, 4, 8, 0, 8, 0, 0, 0)$	22
550	001	1	10	$(1, 2, 2, -5, 4, -7, -7, 10)$	11
	011	1	10	$(1, 1, -5, 4, -5, 4, 10, -8)$	22
	011	2	10	$(2, -10, -2, 3, -2, 3, 5, 1)$	42
	111	1	10	$(-3, 5, 5, -8, 5, -8, -8, 10)$	1
	111	2	10	$(2, -10, -10, 9, -10, 9, 9, 1)$	105
703	001	1	10	$(1, 2, 2, -5, 4, -7, -7, 10)$	1
	001	2	10	$(-1, -4, -4, -5, -6, 6, 6, 10)$	4
	011	2	10	$(1, -10, -2, 5, -2, 5, 5, -2)$	14
	111	2	10	$(2, -10, -10, 9, -10, 9, 9, 0)$	2
1344	001	1	30	$(30, 17, 17, 0, 24, 0, 0, 0)$	11
	011	2	30	$(21, 7, 20, 30, 20, 30, 30, 0)$	9
	011	1	30	$(20, -7, 30, 0, 30, 0, 0, 0)$	2
	011	1	30	$(21, -6, 30, 0, 30, 0, 0, 0)$	4
	001	1	30	$(30, 17, 17, 0, 25, 0, 0, 0)$	1

Substitute each displayed f_i into the scalar coefficient formula of Section 2, and then substitute the exact counting parametrization $N = N(\xi)$. Rational arithmetic gives

$$\sum_i \frac{t_i}{|\operatorname{Aut} H_i|} S(H_i, \ell_i, f_i; N(\xi)) = -b_0 - \sum_{j=1}^{4} b_j \xi_j, \tag{3.2}$$

where all five b_j are strictly positive; their values are in Appendix B, Tables 5 and 6. Every summand on the left is nonnegative for actual graph counts, whereas the right side is strictly negative.

Moving the count terms to the left and dividing by b_0 gives the certificate of Theorem 2.3 with $\varepsilon = 1$, $\eta_i = t_i/(b_0|\operatorname{Aut} H_i|)$, $z_{F_j} = b_j/b_0$ for $1 \le j \le 4$, and all other $z_F = 0$.

The contradiction for each parameter set. For each v-block, abbreviate

$$E_{v,i} = \frac{S(H_i, \ell_i, f_i; N)}{|\operatorname{Aut} H_i|} \ge 0.$$

The positivity holds for every hypothetical graph with the indicated parameters. Write $b_{v,j}$ for the corresponding entries of Tables 5–6. The four instances of (3.2) now give the following separate numerical contradictions, using only $\xi_j \ge 0$. For $(540, 147, 66, 30)$,

$$\begin{aligned} 0 &\le 102E_{540,1} + 80E_{540,2} + 61E_{540,3} + 37E_{540,4} + 22E_{540,5} \\ &= -b_{540,0} - \sum_{j=1}^{4} b_{540,j}\xi_j \le -\frac{574630684892568}{6496825} < 0. \end{aligned} \tag{3.3}$$

For $(550, 162, 75, 36)$,

$$\begin{aligned} 0 &\le 11E_{550,1} + 22E_{550,2} + 42E_{550,3} + E_{550,4} + 105E_{550,5} \\ &\le -\frac{3153820063124750}{4826151} < 0. \end{aligned} \tag{3.4}$$

For $(703, 182, 81, 35)$,

$$\begin{aligned} 0 &\le E_{703,1} + 4E_{703,2} + 14E_{703,3} + 2E_{703,4} \\ &\le -\frac{827108257923964177}{2718487200} < 0. \end{aligned} \tag{3.5}$$

For $(1344, 221, 88, 26)$,

$$\begin{aligned} 0 &\le 11E_{1344,1} + 9E_{1344,2} + 2E_{1344,3} + 4E_{1344,4} + E_{1344,5} \\ &\le -\frac{11861558480040192}{18036575} < 0. \end{aligned} \tag{3.6}$$

These four certificates use only degrees 1 and 2. After the counting substitutions, the displayed positive combination has a strictly negative constant term and nonpositive coefficients on the four remaining nonnegative counts.

3.2 The six-point cases 351 and 476

For $(351, 140, 73, 44)$ and $(476, 133, 60, 28)$ we retain counts through $p = 6$ and the degree-one and degree-two kernels on three and four reference vertices. In these two cases, respectively, $3\max(|\alpha|, |\beta|) = 24/35$ and $15/19$, both less than 1. Thus every reference Gram matrix of order at most four is positive definite by strict diagonal dominance. This is the finite combination of the constraints of Section 2 for $m = 3, 4$. There are 208 graph types, counting rank 185, and 23 free counts. No extra local exclusion or integrality condition is used.

Let $S_{v,i} = S(H_i, \ell_i, f_i)$, where the rational functions are listed in Appendix C. Appendix D specifies positive integers $L_v, \omega_{v,i}, \zeta_{v,j}$ and the count graphs $G_{v,j}$. Put

$$\eta_{v,i} = \frac{\omega_{v,i}}{vL_v}, \qquad z_{v,j} = \frac{\zeta_{v,j}}{vL_v}.$$

Expanding (2.16) and applying the counting equations gives, for $(351, 140, 73, 44)$,

$$0 \le \sum_{i=1}^{17} \eta_{351,i} S_{351,i} + \sum_{j=1}^{7} z_{351,j} N_{G_{351,j}} = -1 < 0, \tag{3.7}$$

and, for $(476, 133, 60, 28)$,

$$0 \le \sum_{i=1}^{16} \eta_{476,i} S_{476,i} + \sum_{j=1}^{8} z_{476,j} N_{G_{476,j}} = -1 < 0. \tag{3.8}$$

These are two instances of Theorem 2.3 with $\varepsilon = 1$.

3.3 The seven-point certificate for (405,132,63,33)

For these parameters the restricted eigenvalues are $33, -3$, the embedding dimension is $d = 30$, and the full spherical Gram matrix is

$$G = \frac{9I + 3\,\mathrm{Adj}(\Gamma) - J}{8}, \qquad \alpha = -\frac{1}{8}, \quad \beta = \frac{1}{4}.$$

The constraints of Section 2 exclude this case with counting order $p = 7$, at most five reference vertices, and degrees $\ell = 0, 1, 2$.

Proposition 3.1. *There is no strongly regular graph with parameters* $(405, 132, 63, 33)$. *An exact certificate uses* 67 *nonnegative count terms and* 107 *weighted kernel sums:* 54 *of degree zero,* 50 *of degree one, and three of degree two.*

Exact certificate. For an integer $h \ge 0$, write $(v)_h = v(v-1)\cdots(v-h+1)$, with $(v)_0 = 1$. Use the positive rescaling

$$y_F = \frac{2^{\binom{h}{2}} N_F}{(405)_h}, \qquad h = |F|.$$

All 1252 graph types through order seven are retained, with $y_{K_1} = 1$. The 4135 ordinary one-vertex extension equations admit 1078 triangular elimination steps, leaving 173 free scaled graph counts and the constant coordinate. Write $y = Px$, with $y \in \mathbb{R}^{1252}$, $x = (1, x_1, \ldots, x_{173})^T$, and $P \in \mathbb{Q}^{1252\times 174}$, where $x_j = y_{C_j}$, $1 \le j \le 173$, for the free graph types. The supplied rational matrix P is checked against every extension equation, and its free rows form an identity matrix.

For an anchor H of order m, the matrix used in the stored certificate is the positive multiple

$$\widehat{M}_{H,\ell}(y) = \frac{2^{\binom{m+1}{2}}}{(405)_{m+1}} M_{H,\ell}(N).$$

This normalization follows directly by separating diagonal and distinct-vertex terms in the definition of $S(H, \ell, f)$; the ratio of the pair and single-vertex factors is $(404 - m)/2^{m+1}$. Each selected anchor Gram matrix is positive definite. The integer directions are defined on all 2^m profiles; the three degree-two energies use $H = K_{2,3}$.

Let $R_i(y) \le 0$ be the 174 selected sources, numbered as in Supplement [S405]: the first 67 are $-y_F$, and the remaining 107 are $-f_i^T \widehat{M}_{H_i,\ell_i}(y) f_i$. The kernel functions retain the source indices $i = 68, \ldots, 174$. Every direction f_i and every full positive rational coefficient w_i is printed

there, together with formulas that reconstruct all row coefficients using the ordinary counting equations.

For example, source $i = 76$ has $H = K_{2,3}$ with bipartition $\{0,1\} \cup \{2,3,4\}$ and degree $\ell = 2$. Its 32 values, read row by row in binary lexicographic profile order, are

$$\begin{array}{rrrrrrrr} 117610 & -293584 & -293584 & -14274 & -293584 & -14274 & -14274 & 0 \\ -34929 & -18491 & -18491 & 265514 & -18491 & 265514 & 265514 & 52070 \\ -34929 & -18491 & -18491 & 265514 & -18491 & 265514 & 265514 & 52070 \\ 0 & -13820 & -13820 & -290273 & -13820 & -290273 & -290273 & 201821 \end{array}$$

and the exact coefficient of its source inequality is

$$w_{76} = \frac{386746425034437}{536754063954592065429687500000000}.$$

This one source is part of the complete positive combination; its positivity alone is not claimed to give the contradiction.

Applying the stated counting substitutions and collecting coefficients in the full combination gives the exact identity

$$\sum_{i=1}^{174} w_i R_i(Px) = c_0 + \sum_{j=1}^{173} c_j x_j.$$

Here the c_j are local to this combination of nonpositive sources. The empty-set extension equations and count nonnegativity imply $0 \le y_F \le 2^{\binom{|F|}{2}}$ by induction on $|F|$. Put $U_j = 2^{\binom{|C_j|}{2}}$ and $E_{405} = -\sum_{c_j<0} c_j U_j$. Exact rational arithmetic yields the following convenient bounds on the two sides of the residual payment:

$$c_0 > \frac{570}{10^{10}}, \qquad 0 \le E_{405} < \frac{11}{10^{10}}.$$

In particular, adding explicit nonpositive count-bound terms produces

$$\begin{aligned} 0 \ &\ge \sum_{i=1}^{174} w_i R_i(Px) + \sum_{c_j \ge 0} c_j(-x_j) + \sum_{c_j<0} (-c_j)(x_j - U_j) \\ &= c_0 - E_{405} = \Delta_{405} > \frac{559}{10^{10}} > 0. \end{aligned} \tag{3.9}$$

After multiplying by -1, this is Theorem 2.3 with $\varepsilon = \Delta_{405}$ and the proved count-bound inequalities. Supplement [S405] specifies all 174 summands and the counting recursion used to derive c_0 and c_j; the supplementary verifier independently reconstructs all rows and checks both strict rational bounds above. □

The five-point real-count relaxation is feasible at every Gegenbauer degree; Supplement [S405, Section S2] gives an exact witness and proof.

4 Adding structural information for three further exclusions

We now treat

$$(375, 102, 45, 21), \qquad (441, 88, 35, 13), \qquad (575, 112, 45, 16).$$

We add integer common-neighbour inequalities and unique-extension identities to the Gegenbauer positivity and counting information of Section 2. Both are expressed in the same labelled counts N_F. These proved conditions strengthen the same counting system.

4.1 Integer common-neighbour counts

Let H be a labelled graph on three vertices. For an induced embedding $\phi : H \hookrightarrow \Gamma$, let t_ϕ be the number of vertices outside $\phi([3])$ adjacent to all three selected vertices. Write $(t)_j = t(t-1)\cdots(t-j+1)$ and $(t)_0 = 1$, and define

$$T_j(H) = \sum_{\phi: H \hookrightarrow \Gamma} (t_\phi)_j.$$

Let $\mathcal{E}_j(H)$ be the family of labelled graphs obtained from H by adding j new labelled vertices, each adjacent to all of H, with every possible edge pattern among the new vertices. Counting ordered distinct common neighbours gives

$$T_j(H) = \sum_{F \in \mathcal{E}_j(H)} N_F. \tag{4.1}$$

Here $T_0(H) = N_H$; the labels record the ordered neighbours, so no factorial or automorphism factor occurs. Isomorphic extensions retain their full multiplicity.

For integers a, b, the polynomial

$$p_{a,b}(t) = (t-a)(t-a-1)(t-b)(t-b-1) \tag{4.2}$$

is nonnegative at every integer t. Thus $\sum_\phi p_{a,b}(t_\phi) \geq 0$. Expanding in the basis $(t)_0, \ldots, (t)_4$ and using (4.1) gives a linear inequality in counts of graphs on at most seven vertices. For example,

$$p_{1,4}(t) = (t)_4 - 6(t)_3 + 20(t)_2 - 40(t)_1 + 40,$$

so the 375 case uses

$$T_4(\overline{K_3}) - 6T_3(\overline{K_3}) + 20T_2(\overline{K_3}) - 40T_1(\overline{K_3}) + 40T_0(\overline{K_3}) \geq 0.$$

The complete list of quartics with positive multipliers in the final certificates is

v	H	a	b
375	$\overline{K_3}$	1	4
441	$\overline{K_3}$	0	4
441	K_3	18	28
575	$\overline{K_3}$	1	6
575	$\overline{K_3}$	1	7
575	P_3	8	14

Each inequality holds at every reference copy before averaging.

4.2 An integral Gram matrix and its short vectors

The restricted eigenvalues are r and -3, with r of multiplicity d. Put

$$c = r + 3, \qquad \rho = \frac{k+3}{v}, \qquad B = \mathrm{Adj}(\Gamma) + 3I.$$

The standard spherical Gram matrix satisfies $(3-\rho)G = B - \rho J$. Choose a unit vector e_0 perpendicular to the spherical span and set

$$b_x = \sqrt{3-\rho}\, p_x + \sqrt{\rho}\, e_0, \qquad \delta = \sqrt{\rho}\, e_0.$$

Then B is the Gram matrix of the b_x, and

$$\langle b_x, b_y \rangle = \begin{cases} 3 & x = y, \\ 1 & x \sim y, \\ 0 & x \neq y,\ x \not\sim y. \end{cases} \tag{4.3}$$

Moreover $\delta = v^{-1} \sum_x b_x$ and $\langle b_x, \delta \rangle = \|\delta\|^2 = \rho$. The SRG identities give

$$B^2 = cB + \mu J, \qquad K := B - \rho J, \qquad K^2 = cK, \qquad \operatorname{rank} K = d. \tag{4.4}$$

Consider the integer span $\Lambda = \sum_x \mathbb{Z} b_x$. For $q = \sum_x h_x b_x$ with $h_x \in \mathbb{Z}$, define

$$\tau = \sum_x h_x, \qquad Q = \|q\|^2, \qquad z_x = \langle q, b_x \rangle.$$

Since $\langle q, \delta \rangle = \rho\tau$, the value of τ is independent of the representation of q. Equation (4.4) gives

$$\sum_x z_x = (k+3)\tau, \qquad \sum_x z_x^2 = cQ + \mu\tau^2, \qquad \operatorname{Adj}(\Gamma) z = rz + \mu\tau \mathbf{1}. \tag{4.5}$$

Also

$$Q \geq \rho\tau^2, \qquad Q \equiv \tau \pmod 2, \tag{4.6}$$

where the first assertion is Cauchy's inequality with δ and the second follows by expanding the integral Gram form modulo two. Since every z_x is an integer,

$$0 \leq \sum_x z_x(z_x - 1) = cQ + \mu\tau^2 - (k+3)\tau. \tag{4.7}$$

Lemma 4.1. *For each of the three parameter sets in this section, Λ contains no vector of squared norm one or two. Every vector of squared norm three and coefficient sum one equals b_x for a unique vertex x.*

Proof. All three sets satisfy $1/8 < \rho < 1/2$ and $\mu < r - 2$. If $Q = 1$, (4.6) forces $\tau = \pm 1$. Changing sign gives $\tau = 1$, for which the right side of (4.7) is $-2r < 0$. If $Q = 2$, only $\tau = 0, \pm 2$ are possible. For $\tau = 2$ that right side is $2\mu - 4r < 0$, and sign change handles $\tau = -2$.

Suppose that $Q = 2$ and $\tau = 0$. Cauchy's inequality gives $|z_x| \leq \sqrt{6}$. If $z_x = 2$ or -2, respectively $q - b_x$ or $q + b_x$ would have squared norm one. Hence $z_x \in \{-1, 0, 1\}$. By (4.5), the sets $P = \{x : z_x = 1\}$ and $N = \{x : z_x = -1\}$ both have size c. For $p \in P$, the equation $\operatorname{Adj}(\Gamma) z = rz$ gives

$$|\{x \in P \setminus \{p\} : x \not\sim p\}| + |\mathcal{N}_\Gamma(p) \cap N| = 2. \tag{4.8}$$

The analogous equation holds in N. If two vertices of P were nonadjacent, each would have at least r neighbours in P; these would give at least $2r - (c-2) = r - 1 > \mu$ common neighbours. Thus P and N are cliques, and the cross-edges form a bipartite 2-regular graph C, a disjoint union of even cycles.

Let s be the all-one column on $P \cup N$, let e be 1 on P and -1 on N, and set $R = \operatorname{diag}(e)$. Then

$$K_{P \cup N} = R(2I - \operatorname{Adj}(C))R + (1/2 - \rho) s s^T + \tfrac{1}{2} e e^T. \tag{4.9}$$

If C has m cycles, the first term has rank $2c - m$. The vector e is a nonzero kernel vector of that term, so its rank rises by one after adding $ee^T/2$; the other term is PSD. Since every cycle has length at least four,

$$\operatorname{rank} K_{P \cup N} \geq 2c - \lfloor c/2 \rfloor + 1.$$

This is $46 > 34$ for 375 and $54 > 46$ for 575, contradicting (4.4). For 441, the remaining rank-boundary cases are excluded in the next subsection, using only identities already established above.

Once squared norms one and two are excluded, let $Q = 3$, $\tau = 1$. Now $|z_x| \le 3$. A coordinate 3 gives $q = b_x$; a coordinate -3 would give $q = -b_x$, incompatible with $\tau = 1$. A coordinate 2 or -2 would give a squared-norm-two vector. If none of these occur, $z_x \in \{-1, 0, 1\}$ and

$$|\{x : z_x = -1\}| = \frac{3c + \mu - (k+3)}{2} = 3.$$

At a negative coordinate, the last identity in (4.5) says that the sum of neighbouring coordinates is $\mu - r < -2$. But only two other negative coordinates are available, a contradiction. Finally distinct vertices have distinct vectors by (4.3), which proves uniqueness. □

4.3 The rank-boundary argument for 441

Here $c = 28$, $\rho = 13/63$, and $d = 44$. For the centered vectors $u_x = b_x - \delta = \sqrt{3-\rho}\, p_x$, the nonzero eigenvalues of their Gram matrix K are all 28, so

$$\sum_x u_x u_x^T = 28 I_{44}. \tag{4.10}$$

Suppose that the unresolved vector with $Q = 2$, $\tau = 0$ exists. The preceding argument gives cliques P, N of size 28 with cross-edges a union of m even cycles of lengths $2d_1, \dots, 2d_m$, where $d_i \ge 2$ and $\sum_i d_i = 28$. On these 56 vertices put

$$B_0 = 2I + \mathrm{Adj}(C), \qquad s = \mathbf{1}_P + \mathbf{1}_N, \qquad e = \mathbf{1}_P - \mathbf{1}_N, \qquad \beta_0 = \tfrac{1}{2} - \rho.$$

For $H = \Gamma[P \cup N]$, their centered Gram matrix is $K_H = B_0 + \beta_0 s s^T + e e^T/2$. The matrix B_0 is PSD, with one alternating kernel vector per cycle. The vector s belongs to its range and e is a nonzero kernel vector. Hence

$$\operatorname{rank} K_H = 56 - m + 1 \le 44,$$

so only $m = 13$ and $m = 14$ remain.

Since $\tau = 0$, the vector q belongs to the centered spherical span. For an external vertex x, let $a \in \{0,1\}^{56}$ record its neighbours in $P \cup N$. Its neighbour totals in P and N coincide. Indeed, applying (4.10) to the putative vector q gives $\sum_P u_y - \sum_N u_y = 28q$, whose inner product with u_x is zero. The singular Gram-matrix range condition [BV04, Appendix A.5.5] requires $a - \rho s$ to be orthogonal to $\ker K_H$. This kernel consists of combinations of the alternating cycle vectors orthogonal to e. Consequently the neighbour difference between the two sides of cycle i, divided by d_i, is independent of i. Its total is zero, so a is balanced on each cycle and lies in the range of B_0.

Set

$$T = |\mathcal{N}_\Gamma(x) \cap P| = |\mathcal{N}_\Gamma(x) \cap N|, \qquad E = a^T B_0^+ a,$$

where B_0^+ is the inverse on its range and zero on its kernel. The positive eigenvalues of B_0 are at most 4, giving $E \ge \|a\|^2/4 = T/2$. Since $B_0^+ s = s/4$, a rank-one inverse calculation gives the squared residual length after projecting u_x away from the anchor span:

$$\begin{aligned} \gamma_x &= 3 - \rho - E + \rho T - \frac{\rho^2 c}{2} + \frac{\beta_0 (T - c\rho)^2}{4(1 + \beta_0 c/2)} \\ &= \frac{37T^2 + 104T + 6896}{2576} - E. \end{aligned} \tag{4.11}$$

The $ee^T/2$ term contributes nothing because the profile is balanced.

If $m = 13$, the cycles are twelve C_4 and one C_8, or eleven C_4 and two C_6. Their span has dimension 44, so $\gamma_x = 0$. The complete possible energies for a balanced Boolean profile on a single cycle are

cycle	profiles	energies
C_4	6	$0,\ 3/4,\ 1$
C_6	20	$0,\ 5/6,\ 4/3,\ 3/2,\ 2$
C_8	70	$0,\ 7/8,\ 3/2,\ 15/8,\ 2,\ 23/8,\ 7/2$

These rational values follow by inverting $B_0 + ww^T$ on the single cycle, where w is its alternating column, and evaluating all balanced Boolean columns. Thus the total E belongs to $\frac{1}{8}\mathbb{Z}$ or $\frac{1}{12}\mathbb{Z}$. As $2576 = 16 \cdot 7 \cdot 23$, (4.11) forces $37T^2 + 104T + 6896$ to vanish modulo 7 and 23. But

$$37T^2 + 104T + 6896 \equiv 2(T-2)^2 \pmod 7, \qquad 37T^2 + 104T + 6896 \equiv 14(T-16)^2 \pmod{23}.$$

For $0 \le T \le 28$ this forces $T = 16$, and hence $E = 7 < T/2$, a contradiction.

If $m = 14$, all cycles are C_4 and their span has dimension 43. Let n_1 count cycles where the profile selects one vertex from each side, and n_2 those where it selects all four vertices. Then

$$n_1, n_2 \ge 0, \quad n_1 + n_2 \le 14, \qquad T = n_1 + 2n_2, \qquad E = \tfrac{3}{4}n_1 + n_2.$$

Substitution in (4.11), over these 120 integer pairs, gives exactly the following possibilities with $\gamma_x \ge 0$:

n_2	n_1
0	$0, 1, 2, 3, 4$
1	$0, 1, 2$
2	$0, 1$
3	0
13	$0, 1$
14	0

In each of these 14 cases direct rational substitution gives

$$\gamma_x \ge \frac{11}{161} + \frac{237}{8372} n_2(x). \tag{4.12}$$

On each C_4, the two vertices in P have 28 common neighbours inside $P \cup N$, hence $35 - 28 = 7$ outside. The balance condition on that cycle makes these precisely its contributions to n_2. Consequently $\sum_{x \notin P \cup N} n_2(x) = 14 \cdot 7 = 98$. Summing (4.12) over the 385 external vertices yields

$$\sum_{x \notin P \cup N} \gamma_x \ge 385 \frac{11}{161} + 98 \frac{237}{8372} = \frac{17389}{598} > 28.$$

Projection of (4.10) onto the one-dimensional orthogonal complement of the anchor span instead gives $\sum_x \gamma_x = 28$. The anchor residuals vanish, so this is the required contradiction and completes the boundary step in Lemma 4.1.

4.4 Unique extensions and new equations in the counts

For a labelled graph H, put $B_H = 3I + \mathrm{Adj}(H)$. By Lemma 4.1, an integer column h with

$$\mathbf{1}^T h = 1, \qquad h^T B_H h = 3, \qquad B_H h \in \{0,1\}^{|H|} \tag{4.13}$$

forces a unique external vertex at every induced embedding of H. Indeed, the vector sum is a vertex vector; its inner products with the selected vertices give exactly the indicated adjacency profile, and none is three, so it is external. A second vertex with that profile has inner product

$h^T B_H h = 3$ with the same vector sum and must coincide with it. If F is this labelled one-vertex extension, restriction and completion are inverse bijections, so

$$N_H = N_F. \tag{4.14}$$

Likewise an integer column with $h^T B_H h \in \{1, 2\}$ proves $N_H = 0$.

For example, suppose x_1, x_2 and y_1, y_2, y_3 induce $K_{2,3}$. The vector

$$b_{y_1} + b_{y_2} + b_{y_3} - b_{x_1} - b_{x_2}$$

has coefficient sum one and squared norm $15-12 = 3$. It is the unique missing vertex completing this copy to $K_{3,3}$. With the labelled-count convention of Section 2, the resulting identity is simply $N_{K_{2,3}} = N_{K_{3,3}}$.

One can obtain an equation on seven-point counts even when the unique extension has eight vertices. Suppose an eight-vertex graph F has an integer column a satisfying

$$B_F a = 0, \qquad \mathbf{1}^T a = 0.$$

For an index i with $a_i = \pm 1$, put $H_i = F - i$ and $h^{(i)} = -a_i (a_j)_{j \neq i}$. The displayed null relation shows that $h^{(i)}$ satisfies (4.13), with profile equal to that of the deleted vertex. Hence, if also $a_j = \pm 1$,

$$N_{H_i} = N_F = N_{H_j}. \tag{4.15}$$

Eliminating N_F leaves an equation involving only seven-vertex counts.

4.5 The three exact contradiction certificates

The source inequalities are count nonnegativity, the degree $0, 1, 2$ kernel sums $S(H, \ell, f)$, and the six quartic inequalities above. The zero-count and completion equations follow from Lemma 4.1 and (4.13). The rational source lists are identified in Appendix A.2; their conversion to the notation of Section 2 is given in Appendix A.4.

Use the positive scaling

$$y_F = \frac{2^{\binom{h}{2}} N_F}{(v)_h}, \qquad h = |F|, \qquad (v)_h = v(v-1)\cdots(v-h+1). \tag{4.16}$$

Thus $y_{K_1} = 1$. Ordinary one-vertex equations eliminate a vertex of degree at most two by

$$N_{H_B} = e(H, B) N_H - \sum_{B \subsetneq B' \subseteq V(H)} N_{H_{B'}}, \qquad |B| \leq 2.$$

Each substitution decreases the number of vertices or increases the number of edges at fixed order. Through order seven, it leaves the singleton and 173 graph types of minimum degree at least three. The proved zero-count and completion equations then have rank 63, leaving 110 free scaled graph counts $x_j = y_{C_j}$ and the constant 1. All substitutions are rational consequences of the displayed equations.

Every C_j is connected: two components of minimum degree three would require at least eight vertices. Choosing a spanning tree gives

$$N_{C_j} \leq v k^{h-1}, \qquad 0 \leq x_j \leq \frac{2^{\binom{h}{2}} k^{h-1}}{(v-1)_{h-1}} < U_v, \qquad h = |C_j|.$$

For $4 \leq h \leq 7$, direct substitution permits $U_{375} = 1024$ and $U_{441} = U_{575} = 256$.

After these substitutions, write the selected source inequalities as $R_i(x) \leq 0$. Their positive rational combination is

$$R_v(x) = \sum_i w_i R_i(x) = c_{v,0} + \sum_{j=1}^{110} c_{v,j} x_j, \qquad w_i \in \mathbb{Q}_{>0}.$$

Adding the indicated nonpositive count-bound terms cancels every remaining variable:

$$\begin{aligned}\mathcal{L}_v(x) &:= R_v(x) + \sum_{c_{v,j}\geq 0} c_{v,j}(-x_j) + \sum_{c_{v,j}<0} (-c_{v,j})(x_j - U_v)\\ &= c_{v,0} + U_v \sum_{c_{v,j}<0} c_{v,j} =: \Delta_v.\end{aligned} \tag{4.17}$$

Every term on the first line is nonpositive for a hypothetical graph. The exact source combinations yield the three contradictions

$$\begin{aligned}0 \geq \mathcal{L}_{375}(x) = \Delta_{375} &> \frac{7}{10^9} > 0,\\ 0 \geq \mathcal{L}_{441}(x) = \Delta_{441} &> \frac{9}{10^7} > 0,\\ 0 \geq \mathcal{L}_{575}(x) = \Delta_{575} &> \frac{3}{10^6} > 0.\end{aligned} \tag{4.18}$$

Negating (4.17) gives the extended certificate in Theorem 2.3, with $\varepsilon = \Delta_v$. Thus none of the three parameter sets in this section is realizable. The source inventories are recorded in Appendix A.4. Only finite rational sums and comparisons are needed to evaluate these certificates; the search that produced them is not part of the proof.

A Exact evaluation and certificate data

A.1 Polynomial evaluation and singular references

We use the normalized Gegenbauer recurrence in [M08, Section 2.1]:

$$P_0^{(D)}(t) = 1, \qquad P_1^{(D)}(t) = t, \qquad P_\ell^{(D)}(t) = \frac{(2\ell + D - 4)tP_{\ell-1}^{(D)}(t) - (\ell - 1)P_{\ell-2}^{(D)}(t)}{\ell + D - 3}, \quad \ell \geq 2.$$

For $D = 2$ this is the Chebyshev recurrence. At residual inner product z and squared residual lengths η, η', the homogeneous kernels satisfy

$$K_0 = 1, \qquad K_1 = z, \qquad K_\ell = \frac{(2\ell + D - 4)zK_{\ell-1} - (\ell - 1)\eta\eta' K_{\ell-2}}{\ell + D - 3}, \quad \ell \geq 2.$$

Their coefficients are rational when the spherical data and the inverse reference Gram matrix are rational. The diagonal value is 1 at degree zero and η^ℓ at positive degree, including $\eta = 0$.

Every anchor used for the kernels in Section 3 has a positive definite Gram matrix. For a singular anchor, projection would instead use its actual span, residual dimension $d - \operatorname{rank} G_H$, and the corresponding range conditions. The singular configurations in Section 4 are treated explicitly by their separate structural arguments.

A.2 Certificate snapshot and verification

The public companion repository is

`https://github.com/SelenaJXWANG/srg-gegenbauer-counting`

For reproducibility, the certificate snapshot dated `2026-10-14` is fixed at commit `85cd39b` [Data]. Readers may download that snapshot using GitHub's source archive or check it out with Git. This manuscript revision adds repository references without altering the certificate data or printed numerical tables. The repository's `paper/` directory contains the article and Supplement [S405], in PDF and TeX. The fixed-input manifest `extras/certificate_inputs.sha256` lists the mathematical inputs and exact verification engines. Its SHA-256 is

58ebb057e5d840a1571d503bf2c3a4fc0d1854fe6c7336615eb1fc7bc0398b9b

The manifest excludes the manuscript and generated reports, so this identifier does not depend on the document that prints it. The complete release file hashes, including the papers, are in `extras/SHA256SUMS`.

Table 3 locates the certificate used for each parameter set. A case directory is `results/certificates/v_k_lambda_mu/`, with the four numbers substituted for its name. The exact paths, supporting counting and structural source files, and the corresponding verification engines are indexed in `extras/certificate_index.json`. The complete data are in the `results/certificates/` directory.

Table 3: Certificate inputs used for the displayed contradictions. Filenames are relative to the indicated case directory; the four five-point certificates are read directly from the main TeX tables.

(v, k, λ, μ)	Certificate source
$(351, 140, 73, 44)$	`raw_certificate.json`
$(375, 102, 45, 21)$	`exact_certificate.json`
$(405, 132, 63, 33)$	`ordinary_counting_certificate/exact_certificate.json`
$(441, 88, 35, 13)$	`exact_certificate.json`
$(476, 133, 60, 28)$	`raw_certificate.json`
$(540, 147, 66, 30)$	Tables 2 and 5
$(550, 162, 75, 36)$	Tables 2 and 6
$(575, 112, 45, 16)$	`exact_certificate.json`
$(703, 182, 81, 35)$	Tables 2 and 6
$(1344, 221, 88, 26)$	Tables 2 and 5

To reproduce the fixed snapshot, run

```
git clone https://github.com/SelenaJXWANG/srg-gegenbauer-counting.git
cd srg-gegenbauer-counting
git checkout 85cd39b74c9ef8a61ab746deb1102af3eeae7e10
python3 -S code/verify.py
```

This requires Python 3.9 or later and only its standard library. The five stages check the retained ten-case constructions, the ordinary counting proof and all-degree comparison for 405, the actual short TeX certificates, the expanded TeX certificates, and the article's parameter and structural data. A successful run ends with

```
PASS: five exact verification stages; fixed inputs unchanged.
```

The generated report is `results/latest_verification/summary.json`; the retained check for this release is `results/release_checks/summary.json`. No SDP solver is needed. The common command also stages the preserved engines with their required input paths; individual engine files are not separate public entry points.

The same command is run by the repository's GitHub Actions workflow. For the fixed snapshot above, it completed successfully under Python 3.9 and 3.12; the recorded workflow run provides the execution logs. These checks verify the finite rational calculations, while the counting and positivity arguments in the article justify the mathematical implications. Code is licensed under MIT; the article, supplement, and certificate data are licensed under CC BY 4.0.

Where earlier raw JSON certificates are retained, they need not match the printed five-point directions. The TeX tables are the inputs for those four printed proofs, and 1344 also has an independent reconstruction in `code/verification/verify_1344_certificate.py`. For 405, `ordinary_counting_certificate/` is the present proof; `earlier_structural_certificate/` is a distinct retained proof using extra zero-count relations. Optional numerical searches may produce further certificates and are separate from this replay.

A.3 Exact data and graph encodings

The supplementary data contain all rational source rows, directions, weights, coordinate substitutions and verification programs. The verifiers reconstruct graph types, count substitutions and kernel coefficients, check reference Gram matrices by exact elimination, and check each final rational sign. They require no numerical optimizer. Appendices C and D print the functions and multipliers for 351 and 476; Supplement [S405] prints the complete 405 certificate and its reconstruction formulas. The full numerical source lists for the structural cases accompany the article; Section 4 proves their source conditions and gives the resulting explicit contradictions.

For the five-point and six-point certificate data, edge strings use the order $01, 02, \dots, 0(h-1), 12, \dots, (h-2)(h-1)$. For the structural and 405 data, an integer graph mask puts edge ij, $i<j$, in bit $j(j-1)/2+i$, giving the order $01, 02, 12, 03, 13, 23, \dots$. In the structural data, a catalogue record (id, h, b) identifies id with the h-vertex graph of mask b; catalogue identifiers are not masks. The graphs C_j follow the free-column order recorded with the affine substitution. All profile vectors use binary lexicographic order with coordinate 0 first. The decoding convention is part of the certificate specification.

A.4 Normalization of the structural certificates

For an m-vertex anchor, put $D = d - m$ and $u_x = \sqrt{3-\rho}\, p_x$. Write $\zeta = \langle u_x^\perp, u_y^\perp \rangle$, $\eta = \|u_x^\perp\|^2$ and $\eta' = \|u_y^\perp\|^2$. The structural source data use the kernels 1, ζ and $\zeta^2 - \eta\eta'/D$. Consequently a source row before its positive row normalization is

$$-\frac{2^{\binom{m+1}{2}}}{(v)_{m+1}}\,\chi_\ell S(H,\ell,f), \qquad (\chi_0,\chi_1,\chi_2) = \left(1, 3-\rho, (3-\rho)^2\frac{D-1}{D}\right).$$

The R_i in (4.17) are the resulting rows after the stored positive row normalization. All reference Gram matrices are positive definite. The stored f gives rational values on the retained profiles in binary lexicographic order. A profile is omitted only if its one-vertex extension has a proved zero count or its prescribed residual squared length is negative; neither can occur at an actual reference copy. Set $f = 0$ on every such profile. Coefficients at forbidden graph counts are removed by their zero-count equations. These conventions introduce no additional sign or scale.

For each parameter set, the data contain 58 integer witnesses for zero counts and 273 witnesses for 76 distinct completion equations. After ordinary counting substitutions their rank is 63. The exact affine map has 110 free scaled graph counts and constant coordinate 1; its free rows are an identity matrix. Exact substitution and the rank calculation show that it parametrizes the entire solution space of these equations. The rational source lists, row normalizations and positive weights determine every coefficient in (4.17), including the bounded residual terms.

Table 4: Sources in the three structural certificates, before adding the count-bound terms of (4.17).

v	Count sources	Kernel sources	Quartics	Total
375	13	97	1	111
441	21	88	2	111
575	60	48	3	111

A.5 Five-point and six-point data normalization

The retained five-point and six-point data store the counts $N_F/(v|\operatorname{Aut} F|)$ and the energies $S(H,\ell,f)/(v|\operatorname{Aut} H|)$. Multiplying an archived identity by v restores the present normalization, with automorphism factors absorbed into the multipliers. The definition of S is unchanged.

A.6 A separate numerical reproduction

The accompanying reproduction code implements a search for fresh rational certificates. In a separate research run, it produced verified certificates for all ten parameter sets. The six cases of orders five and six are rebuilt from the parameters; the four seven-point cases use fresh optimization on the previously verified counting and structural models. Their rational certificates and conversions to the present S convention pass exact verification. They are alternative positive combinations; all displayed functions, source inventories and margins in this paper refer to the certificates specified here. The distribution includes these printed certificates and the discovery code; the alternative numerical-run outputs are not needed for their verification and are not included.

The numerical reproduction uses CVXPY [DB16] to model the semidefinite programs and Clarabel [GC24] as its default solver. Array operations use NumPy [H+20], and numerical linear programs use SciPy [V+20] through its HiGHS interface. Exact replay uses Python's rational arithmetic and does not depend on a numerical solver.

B Coefficients for the five-point certificates

The entries below are the strictly positive rational coefficients in (3.2). Together with the functions and multipliers in Table 2, they give every coefficient of the four identities in Section 3.

Table 5: Exact coefficients in (3.2), using labelled counts N_F.

	$v = 540$	$v = 1344$
b_0	$\frac{574630684892568}{6496825}$	$\frac{11861558480040192}{18036575}$
b_1	$\frac{519736215717}{124791014600000}$	$\frac{5233918851}{171401572225}$
b_2	$\frac{66130353}{909555500}$	$\frac{123153959}{15944332300}$
b_3	$\frac{2901906377541}{62395507300000}$	$\frac{22833078929}{1028409433350}$
b_4	$\frac{341895786273}{24958202920000}$	$\frac{118254493}{16587248925}$

Table 6: Exact coefficients in (3.2), using labelled counts N_F.

	$v = 550$	$v = 703$
b_0	$\frac{3153820063124750}{4826151}$	$\frac{827108257923964177}{2718487200}$
b_1	$\frac{2768645749393}{8561109258900}$	$\frac{37006789818385019}{667991024363520000}$
b_2	$\frac{19500242539}{15732075150}$	$\frac{133712632548389}{343508042592000}$
b_3	$\frac{296782302161}{23543050461975}$	$\frac{665498801527621007}{8683883316725760000}$
b_4	$\frac{1327509707}{778282659900}$	$\frac{509997958311169}{108548541459072000}$

C Rational directions for the six-point certificates

The tables give all rational directions for 351 and 476. For $m = 4$, the sixteen entries are in binary lexicographic order $0000, 0001, \ldots, 1111$; edge strings list $01, 02, 03, 12, 13, 23$. As before, the vector in a row is a complete specification of the function on adjacency sets.

Table 7: All rational directions for $v = 351$; divide each displayed vector by 30.

i	H_i	ℓ_i	$30f_i$
1	000	1	$(1, 9, 9, -20, 9, -20, -20, 30)$
2	000000	1	$(0, 3, -5, 7, -3, 0, 27, -30, 5, -27, 0, 18, -7, 29, -17, 0)$
3	000001	1	$(0, -7, -7, -5, -9, 5, 5, 16, -9, 5, 5, 16, 15, -1, -1, -30)$
4	000011	1	$(-2, 4, -9, -7, -9, -7, 25, 5, 1, -20, 12, 20, 12, 20, -30, -17)$
5	000011	1	$(0, 0, 11, -18, -11, 18, 0, 0, 0, 0, -24, 30, 24, -30, 0, 0)$
6	000011	2	$(5, -18, 5, -6, 5, -6, 9, 8, 4, 16, -12, 12, -12, 12, -30, 9)$
7	000111	1	$(5, -5, -5, -6, -5, -6, -6, 29, -24, 17, 17, 1, 17, 1, 1, -30)$
8	001011	1	$(8, -30, 4, 15, 4, 15, -8, -9, 4, 15, -8, -9, -8, -9, 3, 12)$
9	001100	1	$(0, 6, -6, 0, -6, 0, 30, -24, 6, -30, 0, 24, 0, 24, -24, 0)$
10	001100	1	$(-6, 9, 9, -6, 9, -6, -1, -9, 9, -1, -6, -9, -6, -9, -9, 30)$
11	001100	2	$(0, 18, -15, -1, 15, -14, 0, -30, -18, 0, 14, 25, 1, -25, 30, 0)$
12	001111	1	$(10, -19, -5, 16, -5, 16, -13, -1, -5, 13, -7, -4, -7, -4, 30, -16)$
13	011110	1	$(15, -8, -8, -2, -8, -1, -1, 14, -8, -1, -1, 14, -2, 14, 14, -30)$
14	011110	2	$(3, -6, -6, 30, -6, -19, -19, 4, -6, -19, -19, 4, 30, 4, 4, 19)$
15	011111	1	$(1, -12, -12, 30, 0, 3, 3, -13, 0, 3, 3, -13, 7, -2, -2, 5)$
16	011111	1	$(-15, 7, 7, -3, 21, -1, -1, -14, 21, -1, -1, -14, -30, -2, -2, 30)$
17	111111	1	$(16, -12, -12, 3, -12, 3, 3, 11, -12, 3, 3, 11, 3, 11, 11, -30)$

Table 8: All rational directions for $v = 476$; divide each displayed vector by 10.

i	H_i	ℓ_i	$10f_i$
1	111	1	$(-2, 5, 5, -7, 5, -7, -7, 10)$
2	000011	1	$(0, 0, 1, -4, -1, 4, 0, 0, 0, 0, -7, 10, 7, -10, 0, 0)$
3	001011	1	$(0, 0, 4, -1, -2, 1, -6, 4, -2, 1, -4, 3, 10, -7, 0, 0)$
4	001011	1	$(4, -10, -1, 7, -1, 7, 0, -4, -1, 7, 0, -4, 0, -4, 0, 3)$
5	001100	1	$(0, -1, 1, 0, 1, 0, 9, -10, -1, -9, 0, 10, 0, 10, -10, 0)$
6	001111	1	$(0, -9, -4, 10, -4, 10, -4, 0, -4, 7, 8, -9, 8, -9, -2, 1)$
7	011110	1	$(0, 9, 9, 1, -9, 0, 0, -10, -9, 0, 0, -10, -1, 10, 10, 0)$
8	011110	1	$(0, 1, -1, 0, -1, -1, 10, -8, 1, -10, 1, 8, 0, 10, -10, 0)$
9	011110	2	$(0, 0, 0, -10, 0, 0, 0, -1, 0, 0, 0, -1, 10, 1, 1, 0)$
10	011110	2	$(0, -6, 6, 0, 10, -2, -7, 7, -10, 7, 2, -7, 0, -4, 4, 0)$
11	011111	1	$(1, 0, 0, 1, -3, -1, -1, 3, -3, -1, -1, 3, 2, 5, 5, -10)$
12	011111	1	$(0, 0, 0, 0, 0, -5, 5, 0, 0, -5, 5, 0, 0, 10, -10, 0)$
13	011111	1	$(0, -4, -4, 10, -2, 1, 1, -3, -2, 1, 1, -3, 8, -3, -3, -1)$
14	011111	2	$(0, 0, 0, 0, -6, 3, 3, 10, 6, -3, -3, -10, 0, 0, 0, 0)$
15	111111	1	$(3, 7, 7, -10, 7, -10, -10, 7, 7, -10, -10, 7, -10, 7, 7, 1)$
16	111111	2	$(2, -9, -9, 0, -9, 0, 0, 10, -9, 0, 0, 10, 0, 10, 10, -3)$

D Integer multipliers for the six-point certificates

The multipliers include all automorphism factors and refer to labelled N and unnormalized S. There are 17 kernel terms and 7 count terms for 351, and 16 kernel terms and 8 count terms for 476. With the normalization of Section 3, each identity has constant -1. The machine-readable

counterparts are the two `raw_certificate.json` files in Table 3; the verifier compares them with every integer printed below. The full tables are retained so that the exact certificate is also specified within the paper. Within each v-block write $\omega_i = \omega_{v,i}$ and $\zeta_j = \zeta_{v,j}$. We print integers as

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix}_{40} = a\,10^{80} + b\,10^{40} + c \qquad (0 \le a, b, c < 10^{40}).$$

Graphs use the edge-string convention in Appendix A. The common scales are

$$L_{351} = \begin{bmatrix} 1027119496478042697272387043594858669 11 \\ 7090228324872847275751373679454059820670 \\ 1076412292550227782101992011071927842880 \end{bmatrix}_{40}.$$

$$L_{476} = \begin{bmatrix} 128341483 \\ 438783363989916763126649099679982917 3466 \\ 4104560806157292825403747795632063212800 \end{bmatrix}_{40}.$$

Table 9: Complete integer weights for $v = 351$. The row i selects the function in Appendix C; G_j is the six-vertex graph with the displayed edge string.

i	ω_i	j	G_j	ζ_j
1	$\begin{bmatrix} 1402620268911696250801740763324 93 \\ 4026424322731728880101491450226890524 49 \\ 9108507321249499450249403888842809600000 \end{bmatrix}_{40}$	1	000010001001011	$\begin{bmatrix} 217271222216938324883488183462 29 \\ 587071787611192404723428107837867671947 8 \\ 420223658323114689800169701667903551633 0 \end{bmatrix}_{40}$
2	$\begin{bmatrix} 902728562501449463194298440654 7 \\ 228540745745715809213454966746331850724 0 \\ 552823394177886525637036688621379245600 0 \end{bmatrix}_{40}$	2	000110011011110	$\begin{bmatrix} 45481764748313957097715799248390 8 \\ 452574648336994341533072838743529612307 5 \\ 207934109772479493065208258447205618881 9 \end{bmatrix}_{40}$
3	$\begin{bmatrix} 188304312248630294141452859386598 \\ 242782967729969521276287442169560351922 7 \\ 880292717675550330741510856093512640000 0 \end{bmatrix}_{40}$	3	000110011011111	$\begin{bmatrix} 34768326390306631024824657159858 7 \\ 468571550719497271825216915018507362573 2 \\ 191602513348964851040430356746940343354 6 \end{bmatrix}_{40}$
4	$\begin{bmatrix} 330074577885806894595637185790 15 \\ 519609237802535117036831768629695145010 4 \\ 468498685530680046273966330578991950000 0 \end{bmatrix}_{40}$	4	000110011111110	$\begin{bmatrix} 33699590430295639064277892183356 20 \\ 453003475295532854610466080862311156562 4 \\ 602671239766731922008221522424971703418 4 \end{bmatrix}_{40}$
5	$\begin{bmatrix} 601771974841698269675813455298 2 \\ 187119642248902495501727995401381171357 8 \\ 627758454250065212529165189632965470000 0 \end{bmatrix}_{40}$	5	000110111111000	$\begin{bmatrix} 60902639169028995807880408453550 6 \\ 730413190720412979806118711958608320494 3 \\ 24526326142662571965951116346637706757 9 \end{bmatrix}_{40}$
6	$\begin{bmatrix} 977735209889834034062569312850 66 \\ 183599299495933629207160082214271305286 2 \\ 983305233879178313378708424841716250000 0 \end{bmatrix}_{40}$	6	001110111111000	$\begin{bmatrix} 11630704423068381820899281260167 6 \\ 506873754770751262315517540871775051127 7 \\ 807825904573626857634734377977439758200 4 \end{bmatrix}_{40}$
7	$\begin{bmatrix} 354000977412690024023807297864 38 \\ 328904998553748526548569182279071424586 4 \\ 801279624531380179683285506267763057600 0 \end{bmatrix}_{40}$	7	001110111111001	$\begin{bmatrix} 14448159020546918344569129195544 14 \\ 585269293627532510194640254930289857584 1 \\ 589516716289640706383027132330957775850 4 \end{bmatrix}_{40}$
8	$\begin{bmatrix} 369509214537610886754996110746 551 \\ 214792093911370746700511270349413164634 3 \\ 620294134711261383215992759738549734400 0 \end{bmatrix}_{40}$			
9	$\begin{bmatrix} 616407449610743941152354783581 36 \\ 479087148407760181296352736198706027829 8 \\ 740193801268984693707258416048063987200 0 \end{bmatrix}_{40}$			
10	$\begin{bmatrix} 231445612675255990474046680219 498 \\ 296519261221478713188953808785422230256 2 \\ 397657013219228207153460193602932518400 0 \end{bmatrix}_{40}$			
11	$\begin{bmatrix} 144842136077889191304253439159 37 \\ 314887351493118315282366623538431382528 8 \\ 203761649666892353391187333637013504000 0 \end{bmatrix}_{40}$			
12	$\begin{bmatrix} 899089634309870492770800215884 828 \\ 249607215509471011429960060384782485374 \\ 577017514388907678202605686304901048000 0 \end{bmatrix}_{40}$			
13	$\begin{bmatrix} 161954488799111804444772990608 552 \\ 102855991880571987803449495670565696135 7 \\ 331754569773594165002678349344128278000 0 \end{bmatrix}_{40}$			
14	$\begin{bmatrix} 413358868328204583373213681940 387 \\ 986634523054266004786958316861535846570 1 \\ 931346666608349467598176139131781729750 0 \end{bmatrix}_{40}$			
15	$\begin{bmatrix} 731569953296160853240567886605 01 \\ 975747532324816598714120615853914249585 \\ 557984015645117199938965034728513130000 0 \end{bmatrix}_{40}$			
16	$\begin{bmatrix} 306359992133369510665913881378 707 \\ 719891173245570804678320668199485603589 5 \\ 630938229562764145030319021081889680000 0 \end{bmatrix}_{40}$			
17	$\begin{bmatrix} 390870661200756186921868370513 89 \\ 354477806586018442024995074743274297411 2 \\ 110779033895981232008945346050526839200 0 \end{bmatrix}_{40}$			

Table 10: Complete integer weights for $v = 476$. The row i selects the function in Appendix C; G_j is the six-vertex graph with the displayed edge string.

i	ω_i	j	G_j	ζ_j
1	$\left\lfloor \begin{matrix} 1 \\ 153290387533992083720437399057021203045 \\ 754969650997644526942715507905451969520 \end{matrix} \right\rfloor_{40}$	1	000010011011110	$\left\lfloor \begin{matrix} 26 \\ 983084908305562915024036867862767748110 \\ 939300422042002315701471130799862355236 \end{matrix} \right\rfloor_{40}$
2	$\left\lfloor \begin{matrix} 5 \\ 421163626498389403018644318472886321159 \\ 234861464398603048720636543520782956800 \end{matrix} \right\rfloor_{40}$	2	000110011011111	$\left\lfloor \begin{matrix} 95 \\ 708359988532101295715210362576129268969 \\ 467458007592744763564872725256665863776 \end{matrix} \right\rfloor_{40}$
3	$\left\lfloor \begin{matrix} 8 \\ 770386311975691435167761473344271024643 \\ 657651096006631383710229016592718388400 \end{matrix} \right\rfloor_{40}$	3	000110011111110	$\left\lfloor \begin{matrix} 51 \\ 422891615578313615850638467025161051840 \\ 511052624402346008134027409912586890976 \end{matrix} \right\rfloor_{40}$
4	$\left\lfloor \begin{matrix} 9 \\ 394667042608147452238431525141556001265 \\ 268988472646218301038966331541025079800 \end{matrix} \right\rfloor_{40}$	4	000110111111000	$\left\lfloor \begin{matrix} 74 \\ 725041157636637634824359263198413795552 \\ 862896944613731582314842999929807049602 \end{matrix} \right\rfloor_{40}$
5	$\left\lfloor \begin{matrix} 3 \\ 615785654518659065687888819388354524540 \\ 618577304280512729944834169398591324000 \end{matrix} \right\rfloor_{40}$	5	000110111111010	$\left\lfloor \begin{matrix} 32 \\ 987957389096941694294850631944740612869 \\ 695606688943937145616544523085779136534 \end{matrix} \right\rfloor_{40}$
6	$\left\lfloor \begin{matrix} 24 \\ 900620278716992790130148077200004137107 \\ 879758964792212475437505620793259161600 \end{matrix} \right\rfloor_{40}$	6	001110111111000	$\left\lfloor \begin{matrix} 126 \\ 260343929912531756443047846593463753544 \\ 248053378819080341004013058929739958835 \end{matrix} \right\rfloor_{40}$
7	$\left\lfloor \begin{matrix} 1 \\ 690707263130732125622572270541627184547 \\ 613920464188434398477142797725360671625 \end{matrix} \right\rfloor_{40}$	7	001110111111011	$\left\lfloor \begin{matrix} 169 \\ 317549368525625331378590209773385586111 \\ 875183165568377264664111504225650974669 \end{matrix} \right\rfloor_{40}$
8	$\left\lfloor \begin{matrix} 2 \\ 155968714226492777810180648160255072685 \\ 829613983501575905240400550024194569180 \end{matrix} \right\rfloor_{40}$	8	011111111011110	$\left\lfloor \begin{matrix} 5 \\ 497266169455331480187330481732681382939 \\ 187146646439973452748004889733303337462 \end{matrix} \right\rfloor_{40}$
9	$\left\lfloor \begin{matrix} 71 \\ 153351416833491118350019500295577887629 \\ 511944834028198464716475064226725584032 \end{matrix} \right\rfloor_{40}$			
10	$\left\lfloor \begin{matrix} 37 \\ 171464277753925149367647836370614017532 \\ 933693038383775388614681414813986058910 \end{matrix} \right\rfloor_{40}$			
11	$\left\lfloor \begin{matrix} 7 \\ 445761295247122084856458333238972962300 \\ 807726900038467411896002302158768038360 \end{matrix} \right\rfloor_{40}$			
12	$\left\lfloor \begin{matrix} 23 \\ 463295452950103318511592310413201110236 \\ 823861153939420253770769391115864033749 \end{matrix} \right\rfloor_{40}$			
13	$\left\lfloor \begin{matrix} 120 \\ 757641944614060609082064312311838830345 \\ 266298012796449102025379519228211453200 \end{matrix} \right\rfloor_{40}$			
14	$\left\lfloor \begin{matrix} 1 \\ 701463304004479506919490660943427723153 \\ 277163906177065008763027067573758250560 \end{matrix} \right\rfloor_{40}$			
15	$\left\lfloor \begin{matrix} 2 \\ 544730504390659930124001388907460610494 \\ 111785253232700261742593635566624941040 \end{matrix} \right\rfloor_{40}$			
16	$\left\lfloor \begin{matrix} 4 \\ 300843687238553544641629547252551003629 \\ 743925785085230375651266128896821668480 \end{matrix} \right\rfloor_{40}$			

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